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COM3502/4502/6502
SPEECH PROCESSING

Lecture 19
Cepstral Analysis

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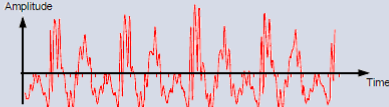
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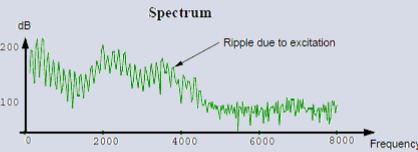
Source-Filter Separation

- 'Cepstral analysis' is another method (*like linear prediction*) for separating the vocal tract filter response from the excitation
- It is based on the observation that the spectrum of a speech signal is the *product* of the excitation spectrum and the vocal tract frequency response

Time-domain speech signal: the vowel "iy"

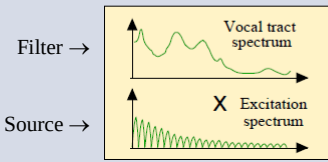


Spectrum



Filter →

Source →



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Source-Filter Separation

- Multiplication can be converted to addition using logs ...

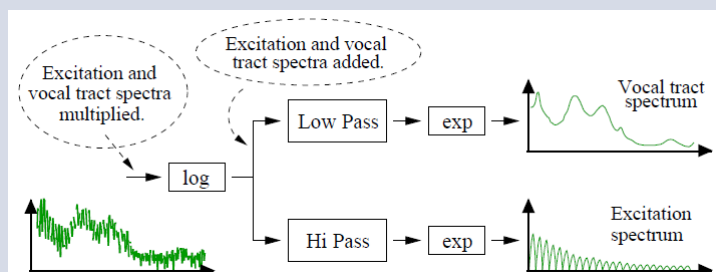
$$\log(a \cdot b) = \log(a) + \log(b)$$

- Hence the '**log spectrum**' can be viewed as the summation of the log excitation spectrum and the log vocal tract frequency response

$$\log(\text{spectrum}) = \log(\text{excitation spectrum}) + \log(\text{vocal tract frequency response})$$

- So, if the log spectrum is treated as if it were a *waveform*, then it can be filtered!
- This approach is called '**homomorphic filtering**'

Homomorphic Filtering



The Cepstrum

- A sequence can be filtered in the time domain or in the frequency domain
- This means that another way of filtering the (*log*) spectrum is to compute its *spectrum*!
- The “spectrum of the (*log*) spectrum” is called the ‘**cepstrum**’ (/kepstrəm/)

$$c_c[m] = \mathcal{F}^{-1} \{ \log \{ \mathcal{F} \{ x[k] \} \} \}$$

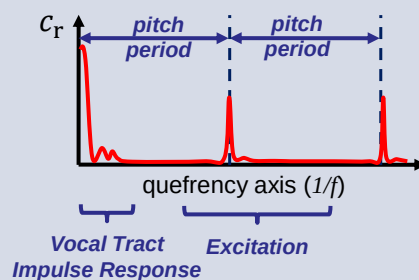
$$= \mathcal{F}^{-1} \{ \log \{ x[n] \} \}$$
- The (often-used real) cepstrum is defined formally as “the *inverse Fourier transform of the log magnitude spectrum of a signal*” ...

$$c_r[m] = \mathcal{F}^{-1} \{ \log |x[n]| \}$$

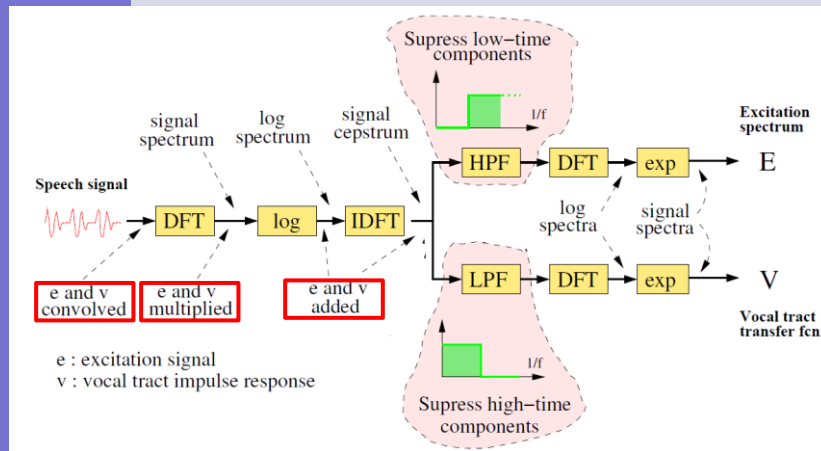
$$= \sum_{k=0}^{L_{\text{DFT}}-1} \log |x[n]| e^{-j2kn\pi/L_{\text{DFT}}}$$

The Cepstrum

- The cepstrum $c_r[m]$ is computed in the ‘**quefrency**’ domain
- Quefrency is measured in units of time (1/f)
- Filtering in the quefrency domain is called ‘**liftering**’



Cepstral Analysis



Applications of Cepstral Analysis

- Pitch estimation
 - based on finding the peak cepstral value in the high quefrency components
- Smoothed spectrum estimation
 - based on taking a DFT of the (*zero-padded*) low quefrency components
 - provides an alternative to the LP spectrum
- Vocoding
 - based on estimating the fundamental frequency and transmitting speech as a sequence of low quefrency cepstral frames
- Automatic speech recognition
 - based on using the low quefrency cepstral components as a representation of the (*smoothed*) spectrum
 - **'Mel Frequency Cepstrum Coefficients'** (*MFCCs*)

Mel-Frequency Cepstrum Coefficients

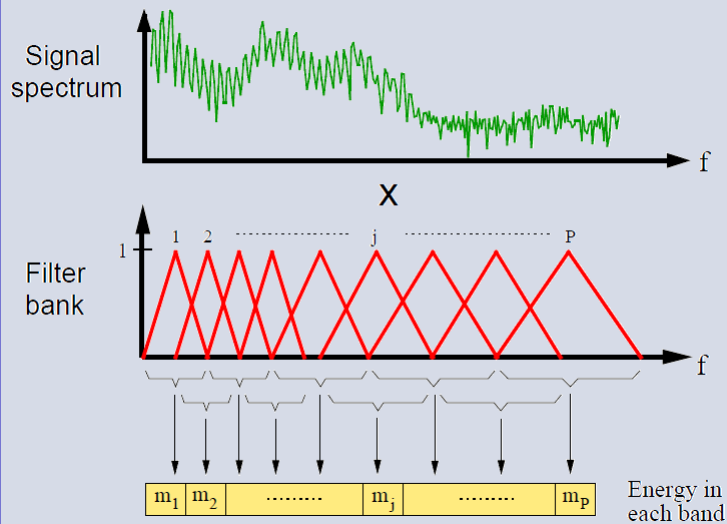
Davis, S. B., & Mermelstein, P. (1980). Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences. *IEEE Trans. Acoustics, Speech, and Signal Processing*, 28(4), 357-366.

- MFCCs are a practical approximation to the full cepstral coefficients ...
 - the frequency scale is non-linear
 - the spectrum is computed using a filter bank
 - the cepstral coefficients are estimated using the '**discrete cosine transform**' (DCT)
- Rather than implement a (*computationally expensive*) bank of digital filters, MFCCs are usually computed using a DFT whose output is grouped into 20-40 bands
- MFCCs have several attractive properties ...
 - they decorrelate the information in the spectrum
 - only a few parameters (10-15) are needed to represent each frame

Which Properties should „good“ Acoustic Features have ?

- All information relevant for interpretation of signal has to be contained in features
 - Enough information necessary for interpretation
 - Not too much (irrelevant) information
 - Efficient calculation
- Good feature generalize well
 - An 'A' of one person should be represented in same manner as 'A' from other person
- Feature vectors X represent speech
- Information about "sub-words" should be well presented

Mel Frequency Filter Bank



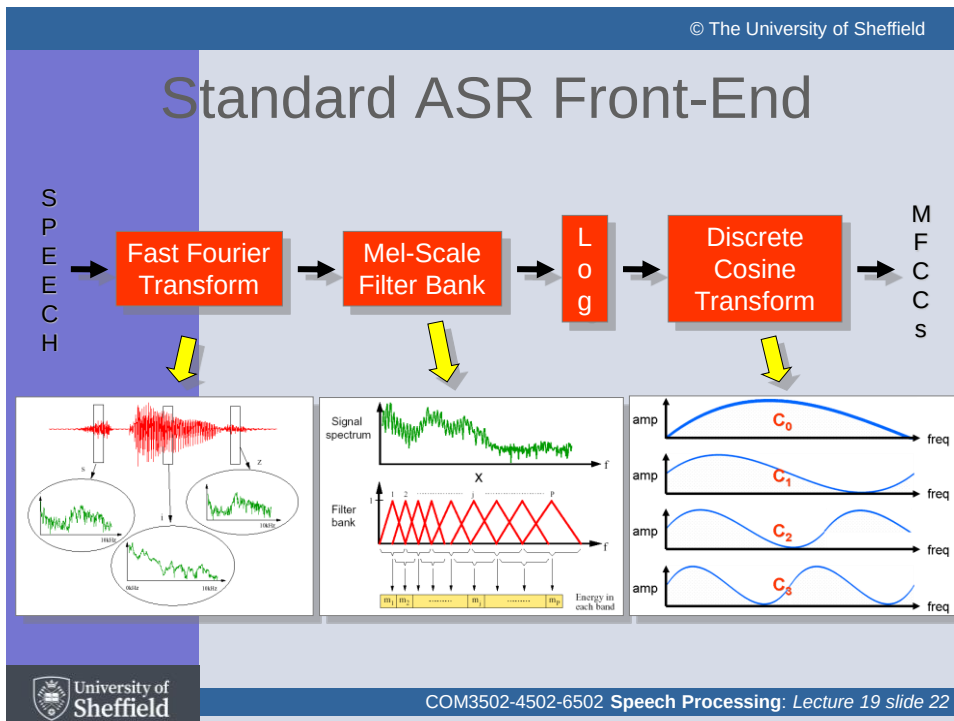
The Discrete Cosine Transform

- The '**DCT**' models a sequence as a sum of cosine components
- It is a *real-valued* approximation to the DFT
- It exploits the fact that the log magnitude spectrum is real-valued, symmetric with respect to 0 and periodic in frequency

$$c_n = \sqrt{\frac{2}{P}} \sum_{i=1}^P m_i \cos \left(\frac{(n + \frac{1}{2})\pi i}{P} \right)$$

... where P is the number of filterbank channels

- Note that c_0 is a measure of the signal energy



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Standard ASR Front-End

- 39-dimensional vector (*frame*)
 - 15-40 msec frame size
 - 10 msec frame shift
 - 100 frames per second
- Each frame contains:
 - 13 MFCCs
 - 13 1st-order derivatives ("deltas")
 - 13 2nd-order derivatives ("delta deltas")
- You can find out how these are used in **'COM4511-6511: Speech Technology'**!

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This slide set has covered ...

- Source-filter separation
- Homomorphic filtering
- The cepstrum
- Cepstral analysis
- The Mel frequency scale
- Mel frequency cepstral coefficients
- A standard ASR front-end

Any Questions ?



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Next ...

Final Overview

/plʌs səm fʌn stʌf/

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